AI-Based Hyper-Local Aviation Weather Forecast Verification System for Sri Lankan Monsoon Patterns

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Weather forecasting plays an important role in aviation safety and operational planning, particularly in regions with rapidly changing atmospheric conditions. Pilots depend on aviation weather reports such as Terminal Aerodrome Forecast (TAF) and METAR to obtain critical atmospheric information before flight operations. These reports provide key meteorological parameters including temperature, wind speed, wind direction, atmospheric pressure (QNH), and relative humidity, which directly influence aircraft performance and flight safety. However, most existing aviation weather forecasting systems rely on large-scale numerical weather prediction models such as the Global Forecast System (GFS) and the European Centre for Medium-Range Weather Forecasts (ECMWF). Although these systems provide reliable global forecasts, they often fail to capture localized atmospheric variations occurring at specific airport locations.

Sri Lanka experiences complex weather conditions influenced by seasonal monsoon systems including the Southwest Monsoon, Northeast Monsoon, and the two inter-monsoon periods. Among these, the Second Inter-Monsoon is characterized by unstable atmospheric conditions, sudden rainfall, and rapid fluctuations in humidity and atmospheric pressure. Existing forecasting approaches typically predict meteorological parameters independently and rarely analyze the interdependent relationship between relative humidity, atmospheric pressure (QNH), and temperature, which may reduce prediction reliability at a local level.

This research proposes a Humidity & QNH Stability Model, a coupled atmospheric prediction approach that learns the relationship between humidity, pressure, and temperature using historical weather datasets collected from Bandaranaike International Airport. Machine learning techniques, including time-series forecasting models, will be applied to analyze atmospheric trends and generate short-term predictions. In addition, the system includes a weather visualization module that presents validated weather data using temperature heatmaps, humidity trend charts, and wind pattern maps to support intuitive interpretation by aviation personnel.

The expected outcome of the research is a predictive and visualization-based aviation weather support system capable of improving localized weather forecasting accuracy and providing reliable weather insights for airport operations. This system aims to support safer flight operations and contribute to improved aviation decision-making within Sri Lanka’s dynamic monsoon climate.

Keywords: Aviation Weather Forecasting, Relative Humidity, QNH Stability Model, Machine Learning, Aviation Meteorology

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TABLE OF CONTENT

Contents

DECLARATION 2

ABSTRACT 3

ACKNOWLEDGEMENT 4

TABLE OF CONTENT 5

LIST OF TABLES 6

LIST OF ABBREVIATIONS 7

1. INTRODUCTION 8

1.1 Background and Context 9

1.2 Literature Review 10

..... 10

1.3 Research Gap 11

2. OBJECTIVE 12

2.1 Main Objective 12

2.2 Specific Objectives 12

3. METHODOLOGY 13

4. HIGH-LEVEL SYSTEM ARCHITECTURE 15

5. USER REQUIREMENTS 17

6. COMMERCIALIZATION PLAN 19

7. BUDGET AND JUSTIFICATION 20

8. WORK BREAKDOWN STRUCTURE(WBS) 20

9. GANTT CHART 23

10. REFERENCES LIST 24

11.1 Survey Instruments 25

11.2 Preliminary UI mockups 25

11.3 Risk Analysis 27

LIST OF TABLES

Table 1- List of Abbreviations 7
Table 2 - Work breakdown table 22
Table 3 - project Implementation Timeline (Gantt Chart) 23
Table 4 - Risk Analysis 27

LIST OF FIGURES

Figure 1- System Diagram 15
Figure 2-System Structure Diagram 16
Figure 3-UI mockup 25
Figure 4-UI Mockup 26
Figure 5-UI Mockup 26

LIST OF ABBREVIATIONS

AI	Artificial Intelligence
ML	Machine Learning
NWP	Numerical Weather Prediction
ANNs	Artificial Neural Networks
SaaS	Software-as-a-Service
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
RMSE	Root Mean Square Error
QNH	Atmospheric Pressure at Mean Sea Level
SDG	Sustainable Development Goal
UI	User Interface
WBS	Work Breakdown Structure
TAF	Terminal Aerodrome Forecast
METAR	Meteorological Aerodrome Report
GFS	Global Forecast System

Table 1- List of Abbreviations

1. INTRODUCTION

Weather forecasting is a critical component of aviation safety, operational planning, and risk management, particularly in regions with complex monsoonal climates such as Sri Lanka. The country experiences four major meteorological periods: the **Northeast Monsoon** (December–February), the **Southwest Monsoon** (May–September), the **First Inter-Monsoon** (March–April), and the **Second Inter-Monsoon** (October–November). Each period is associated with distinct atmospheric patterns, including variations in rainfall, temperature, humidity, and wind conditions, which significantly impact aviation operations [2], [3]. Among these, the **Second Inter-Monsoon** is characterized by rapidly changing weather conditions, including heavy localized rainfall, thunderstorms, and high winds, posing substantial challenges to flight safety and operational planning.

Sudden variations in atmospheric parameters such as relative humidity, atmospheric pressure (QNH), and temperature during the Second Inter-Monsoon can lead to flight delays, diversions, and operational hazards. Accurate and localized forecasts are therefore critical for pilots, air traffic controllers, and airport authorities to make safe and timely operational decisions [2], [6].

Pilots rely on aviation weather reports such as the Terminal Aerodrome Forecast (TAF) and Meteorological Aerodrome Report (METAR) to obtain atmospheric information before aircraft take-off and landing. These reports provide essential meteorological parameters including temperature, wind speed, wind direction, atmospheric pressure, and humidity, which directly influence aircraft performance and airport operations. Accurate weather prediction helps aviation personnel make safe operational decisions and minimize risks related to adverse weather conditions.

Recent developments in **machine learning (ML)** provide effective alternatives for short-term and localized weather forecasting. Techniques such as Artificial Neural Networks (ANNs) and Long Short-Term Memory (LSTM) networks can learn nonlinear temporal patterns from historical data, improving the accuracy of predictions for humidity, pressure, and temperature simultaneously [2], [8]. Multi-variable models have shown better performance than single-variable approaches, particularly for aviation operations that require precise and reliable weather information [8].

Despite these advancements, there is limited research focusing on localized forecasting for the Second Inter-Monsoon period in Sri Lanka. Existing studies rarely utilize airport-specific datasets, such as those from **Bandaranayake International Airport**, and often lack pilot-friendly visualizations and reliability scoring mechanisms [3], [5].
This research addresses these gaps by developing the Humidity & QNH Stability Model, a coupled atmospheric prediction system that forecasts relative humidity and pressure simultaneously. The system will integrate historical weather data, apply machine learning algorithms (ANN and LSTM), and include a reliability scoring mechanism based on pressure stability trends. Additionally, it will provide visual weather representations, such as heatmaps, wind pattern maps, and trend charts, to help pilots and aviation authorities understand and respond to changing weather conditions efficiently. By combining predictive modeling with visualization and reliability assessment, this study aims to provide a localized, accurate, and interpretable weather forecasting solution for Sri Lanka’s aviation sector during the variable monsoon and inter-monsoon periods.

1.1 Background and Context

Weather forecasting plays a critical role in aviation safety and operational decision-making. Accurate atmospheric information is essential for aircraft take-off, landing, and route planning. Pilots and air traffic controllers rely on aviation weather reports such as the Terminal Aerodrome Forecast (TAF) and Meteorological Aerodrome Report (METAR) to obtain key meteorological parameters including temperature, wind speed, wind direction, atmospheric pressure, relative humidity, and visibility. These parameters directly influence aircraft performance, runway conditions, and overall flight safety. Reliable and timely weather predictions therefore play a crucial role in supporting safe aviation operations and efficient airport management.

In Sri Lanka, weather conditions are strongly influenced by monsoon circulation systems and seasonal climate variations. The country experiences four main climatic periods: the Southwest Monsoon, Northeast Monsoon, First Inter-Monsoon, and Second Inter-Monsoon. Among these, the Second Inter-Monsoon period is characterized by highly unstable atmospheric conditions and rapid weather changes. Sudden weather events such as thunderstorms, heavy rainfall, and strong winds frequently occur during this period. Rapid fluctuations in relative humidity, atmospheric pressure (QNH), and temperature can disrupt aviation operations and increase the risk of flight delays, diversions, and operational hazards. These sudden atmospheric changes make it difficult for pilots and aviation authorities to make timely and safe operational decisions.

Several stakeholders are affected by inaccurate or delayed weather predictions. Pilots rely on accurate forecasts to make safe operational decisions during take-off, landing, and route planning. Airlines and airport authorities face operational disruptions and financial losses due to incorrect or delayed weather forecasts. Passengers may experience delays, cancellations, or travel disruptions during unstable weather conditions. In addition, emergency response services depend on reliable weather information to plan and respond effectively to extreme weather situations. Therefore, improving localized weather prediction accuracy is essential for both aviation safety and operational efficiency.

Globally, weather conditions are responsible for a significant proportion of aviation disruptions. Studies indicate that approximately **70–80% of weather-related flight delays are caused by sudden localized weather events**, such as thunderstorms, heavy rainfall, and strong winds. In Sri Lanka, historical aviation data from **Bandaranaike International Airport** shows that sudden rainfall events during the Second Inter-Monsoon period can increase flight disruptions by approximately **25–30%**. These statistics highlight the importance of improving localized weather forecasting systems for aviation environments.

Existing aviation weather forecasting solutions have several limitations. Numerical Weather Prediction (NWP) models are widely used to generate large-scale weather forecasts; however, they often struggle to accurately capture localized atmospheric variations at specific airport terminals and require significant computational resources. In addition, many machine learning-based forecasting approaches predict atmospheric variables independently, such as temperature or rainfall, without considering the interdependent relationships between humidity, pressure, and temperature. This limitation reduces the reliability of predictions during complex weather conditions. Furthermore, many aviation forecasting systems present data in numerical form, which can be difficult for pilots to interpret quickly during operational decision-making.

To address these challenges, this research proposes the development of a **coupled atmospheric prediction model** capable of simultaneously forecasting **relative humidity, atmospheric pressure (QNH), and temperature** using machine learning techniques. The proposed system will utilize historical time-series weather data collected from **Bandaranaike International Airport** and apply preprocessing and data augmentation techniques to improve prediction accuracy, especially for rare extreme weather events. In addition, the system will provide intuitive visualization tools, including **temperature heatmaps, wind pattern maps, and atmospheric trend charts**, to assist pilots and aviation authorities in quickly interpreting weather conditions and making informed operational decisions.

This research also aligns with several global sustainable development goals. It supports **Sustainable Development Goal (SDG) 9: Industry, Innovation, and Infrastructure** by promoting the use of intelligent predictive technologies for aviation systems. Additionally, it contributes to **SDG 13: Climate Action** by supporting improved adaptation to climate-related weather risks through data-driven forecasting approaches. From a research perspective, this study falls within the **Artificial Intelligence and Machine Learning research cluster**, focusing on data-driven decision support systems for aviation safety and real-time operational planning.

1.2 Literature Review

Recent research highlights the growing use of machine learning techniques in weather forecasting, particularly for aviation safety and localized meteorological predictions. Wickramagamage [1] demonstrated the use of **Artificial Neural Networks (ANN)** to forecast monsoon rainfall in Sri Lanka, showing that neural networks can model complex temporal patterns in rainfall data effectively. Similarly, Jayawardena [2] developed a **short-term weather prediction model** for Sri Lanka using machine learning algorithms, emphasizing the ability of data-driven models to improve local forecast accuracy.

Aviation weather hazards, such as thunderstorms and lightning, pose serious operational risks, and several studies have focused on understanding these phenomena for safe flight operations. Wang [3] analyzed common aviation weather hazards in the Yangon Flight Information Region, identifying the critical mechanisms that trigger these events. Jayasinghe [4] investigated lightning activities specifically over **Bandaranaike International Airport**, providing essential insights into localized hazard patterns. Sprenger and colleagues [5] reviewed **high-impact weather events for aviation meteorology**, highlighting the challenges in predicting extreme conditions such as sudden storms, turbulence, and heavy rainfall, and emphasizing the need for accurate and timely forecasting tools.

The use of advanced machine learning frameworks has also gained attention in aviation forecasting. Jeong [6] proposed an **AI-based framework for automated terminal aerodrome forecasting**, combining historical data with predictive models to improve operational decision-making. Shi [7] introduced **Convolutional LSTM networks** for precipitation nowcasting, demonstrating how deep learning models can capture both spatial and temporal correlations in weather data. Chen [8] reviewed several **deep learning approaches for real-time weather forecasting**, showing that neural networks can outperform conventional numerical weather prediction (NWP) models, particularly in short-term predictions.

Furthermore, Bauer and colleagues [9] discussed the **challenges and design choices in global weather and climate models using machine learning**, stressing that accurate modeling requires careful consideration of feature selection, training data, and model generalization. Lerch and Reich [10] examined **postprocessing of ensemble weather forecasts using neural networks**, highlighting how machine learning can improve forecast calibration and reliability. Collectively, these studies demonstrate that multi-variable machine learning models, particularly those combining ANN and LSTM architectures, provide significant advantages in accurately predicting complex and interdependent atmospheric variables such as **humidity, QNH, and temperature**, which is directly relevant to the proposed coupled prediction model for the Second Inter-Monsoon in Sri Lanka.

1.3 Research Gap

Although global weather forecasting models and aviation forecasting systems are widely used in the aviation industry, most of these models are developed using large-scale global meteorological datasets and are not specifically trained using localized airport-level weather observations. As a result, these systems often fail to accurately capture micro-scale atmospheric variations that occur in specific airport environments such as Bandaranaike International Airport in Sri Lanka. Local weather conditions during monsoon and inter-monsoon periods can change rapidly due to interactions between temperature, humidity, and atmospheric pressure, which are not always well represented in global forecasting models.

Another limitation of existing aviation forecasting systems is that many approaches predict atmospheric parameters independently rather than considering the interdependent relationships among variables such as relative humidity, atmospheric pressure (QNH), and temperature. Since these atmospheric parameters influence each other dynamically, predicting them separately may reduce the reliability and accuracy of short-term forecasts, particularly during unstable weather periods such as the Second Inter-Monsoon.

Furthermore, aviation meteorologists and pilots often rely on multiple forecasting systems and manually interpret weather reports such as METAR and TAF. This process can be complex and time-consuming, especially when rapid decision-making is required for flight operations. In addition, many existing aviation forecasting platforms present weather information primarily in numerical formats, which may not provide intuitive insights for pilots who need quick situational awareness during flight planning and airport operations.

Therefore, there is a clear need for a **data-driven, AI-based hyper-local aviation weather forecast verification system** that utilizes localized meteorological data together with advanced machine learning techniques. This research addresses the identified gap by developing relative Humidity & QNH Stability Model, a coupled atmospheric prediction approach that learns the relationships between humidity, pressure (QNH), and temperature using historical weather data from Bandaranaike International Airport. The proposed system also integrates visual weather analytics, including heatmaps and trend charts, to provide simplified and interpretable weather guidance for aviation personnel. By combining predictive modeling with visualization and reliability scoring, the proposed solution aims to improve localized weather forecasting accuracy and support safer aviation decision-making.

2. OBJECTIVE

2.1 Main Objective

To develop a coupled Humidity and QNH prediction model for the Second Inter-Monsoon in Sri Lanka that accurately forecasts short-term atmospheric changes, assigns reliability scores based on pressure stability, and achieves at least 85% trend prediction accuracy within 5 years using historical Bandaranaike International Airport data.

SMART Breakdown:

- **Specific:** Coupled model predicting humidity and QNH with reliability scoring.
- **Measurable:** Achieve ≥85% trend prediction accuracy.
- **Achievable:** Based on historical weather data and machine learning methods (ANN, LSTM).
- **Relevant:** Supports aviation safety and operational decision-making during Second Inter-Monsoon.
- **Time-bound:** Model development, testing, and evaluation completed within **1 year**.

2.2 Specific Objectives

The first objective of this research is to **collect and preprocess historical weather data** from Bandaranaike International Airport for the Second Inter-Monsoon period. This includes gathering records of **relative humidity, atmospheric pressure (QNH), and temperature** and ensuring the dataset is complete, accurate, and free from missing or inconsistent entries. Proper preprocessing will allow the model to learn meaningful patterns and improve prediction reliability.

The second objective focuses on **developing a hybrid data augmentation strategy**. Since extreme weather events are rare but critical for aviation safety, this step involves generating synthetic data and applying time-series augmentation techniques. This approach will improve the model's ability to generalize and handle unusual or extreme atmospheric conditions.

The third objective is to **design and implement a coupled multi-variable machine learning model** that predicts both humidity and QNH simultaneously. By capturing the **interdependencies between atmospheric parameters**, the model will provide more accurate localized forecasts than traditional single-variable prediction models. The chosen architectures will include **Artificial Neural Networks (ANNs) and Long Short-Term Memory (LSTM) networks** to model nonlinear and temporal patterns effectively.

The fourth objective is to **incorporate reliability scoring based on pressure stability**. Each forecast will be assigned a confidence level that quantifies the trustworthiness of predictions. This feature is critical for supporting **pilot and airport authority decision-making**, as it provides actionable information beyond raw predictions.

The fifth objective is to **evaluate the model's performance** using metrics such as **trend accuracy, mean absolute error (MAE), and root mean squared error (RMSE)**. The aim is to ensure the model achieves at least **85% trend prediction accuracy**, demonstrating its operational utility for short-term atmospheric forecasting.

Finally, the sixth objective is to **develop visualization tools** for effective interpretation of forecast data. These tools will include **temperature heatmaps, wind pattern maps, and trend graphs**, enabling pilots and operational staff to quickly understand weather changes and make informed decisions.

3. METHODOLOGY

3.1 Research approach

This research adopts a **Design Science and Experimental research approach** to develop and evaluate a predictive weather forecasting model for the Second Inter-Monsoon period in Sri Lanka. The Design Science approach is used to design and develop a novel artifact, namely the Relative Humidity & QNH Stability Prediction Model, which aims to learn the relationships between atmospheric variables such as humidity, atmospheric pressure (QNH), and temperature.

The experimental approach is applied to test the developed model using historical weather datasets and to evaluate its prediction performance. The study focuses on building a coupled atmospheric prediction model that predicts humidity and pressure simultaneously rather than independently. This approach allows the model to capture the complex interactions between atmospheric variables, thereby improving forecasting accuracy.

3.2 Technical methods and algorithms

The proposed system will utilize time-series data analysis and machine learning techniques to develop the predictive model.

First, time-series preprocessing and analysis will be applied to identify patterns and seasonal trends in weather data during the Second Inter-Monsoon period.

The predictive model will be implemented using Artificial Neural Networks (ANN) and Long Short-Term Memory (LSTM) algorithms. ANN models are capable of capturing nonlinear relationships between atmospheric variables, while LSTM networks are particularly suitable for sequential and time-dependent data such as weather observations.

The model will be designed as a coupled multi-variable prediction system that simultaneously predicts relative humidity and atmospheric pressure (QNH). Additionally, a reliability scoring mechanism based on pressure stability trends will be implemented to evaluate the confidence level of predictions.

3.3 Tools and platforms with justification

The proposed system will be developed using several software tools and platforms.
Python will be used as the primary programming language due to its extensive support for machine learning and data analysis. Libraries such as **Pandas and NumPy** will be used for data preprocessing and numerical computations.
For model development and training, **TensorFlow and Keras** will be used because they provide efficient frameworks for building neural network models.
The development environment will be **Jupyter Notebook or Google Colab**, which allows interactive experimentation, visualization, and easy collaboration.
For weather data visualization, libraries such as **Matplotlib and Seaborn** will be used to generate graphical representations including temperature heatmaps, pressure trend graphs, and wind pattern maps. These visualizations help users interpret weather conditions more easily.

3.4 Data collection and ethical considerations

Historical weather data will be collected from aviation meteorological records associated with **Bandaranaike International Airport**. The dataset will include important atmospheric parameters such as relative humidity, atmospheric pressure (QNH), temperature, wind speed, and wind direction.

The collected data will be organized into time-series format and preprocessed to remove missing or inconsistent values before model training.

Ethical considerations will be maintained during the research process. Permission will be obtained from relevant authorities such as the **Department of Meteorology Sri Lanka** when necessary. The data will be used strictly for academic research purposes, and no sensitive operational information will be disclosed. Proper data storage and management practices will be followed to ensure confidentiality and data integrity.

3.5 Validation strategy and performance metrics

To evaluate the predictive performance of the proposed model, the dataset will be divided into **training, validation, and testing datasets**. This approach ensures that the model can generalize effectively to unseen data.

Cross-validation techniques will also be used to improve model reliability and reduce the risk of overfitting.

Several performance metrics will be used to measure the accuracy of the model:

- **Mean Absolute Error (MAE)** to measure the average difference between predicted and actual values.
- **Root Mean Squared Error (RMSE)** to evaluate prediction accuracy and penalize large prediction errors.
- **Trend Accuracy** to determine how well the model predicts the direction of atmospheric changes.
- **Reliability Score Accuracy** based on pressure stability trends.

These evaluation methods will help determine whether the proposed Humidity & QNH Stability Model provides improved forecasting accuracy compared to conventional weather prediction methods.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

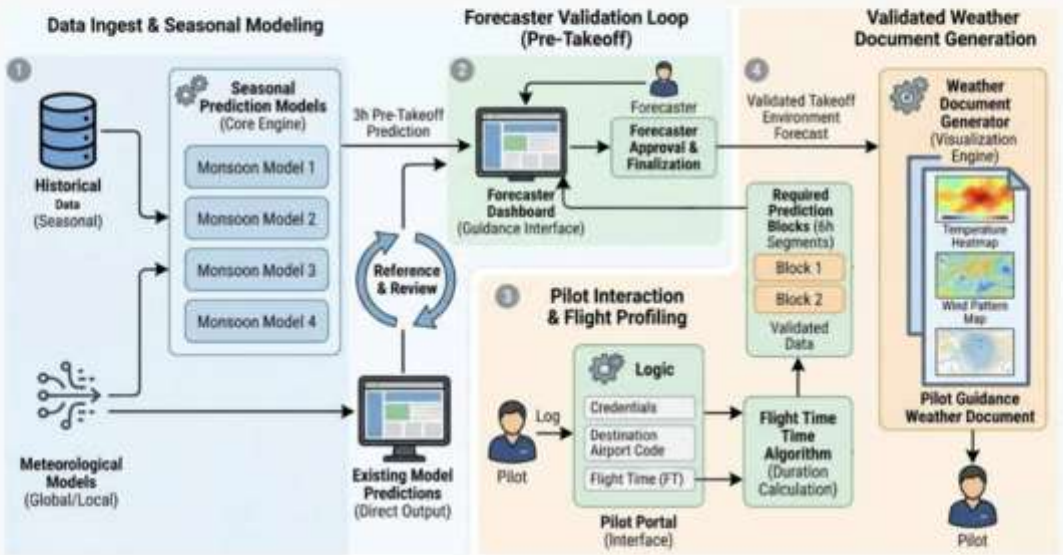


Figure 1- System Diagram

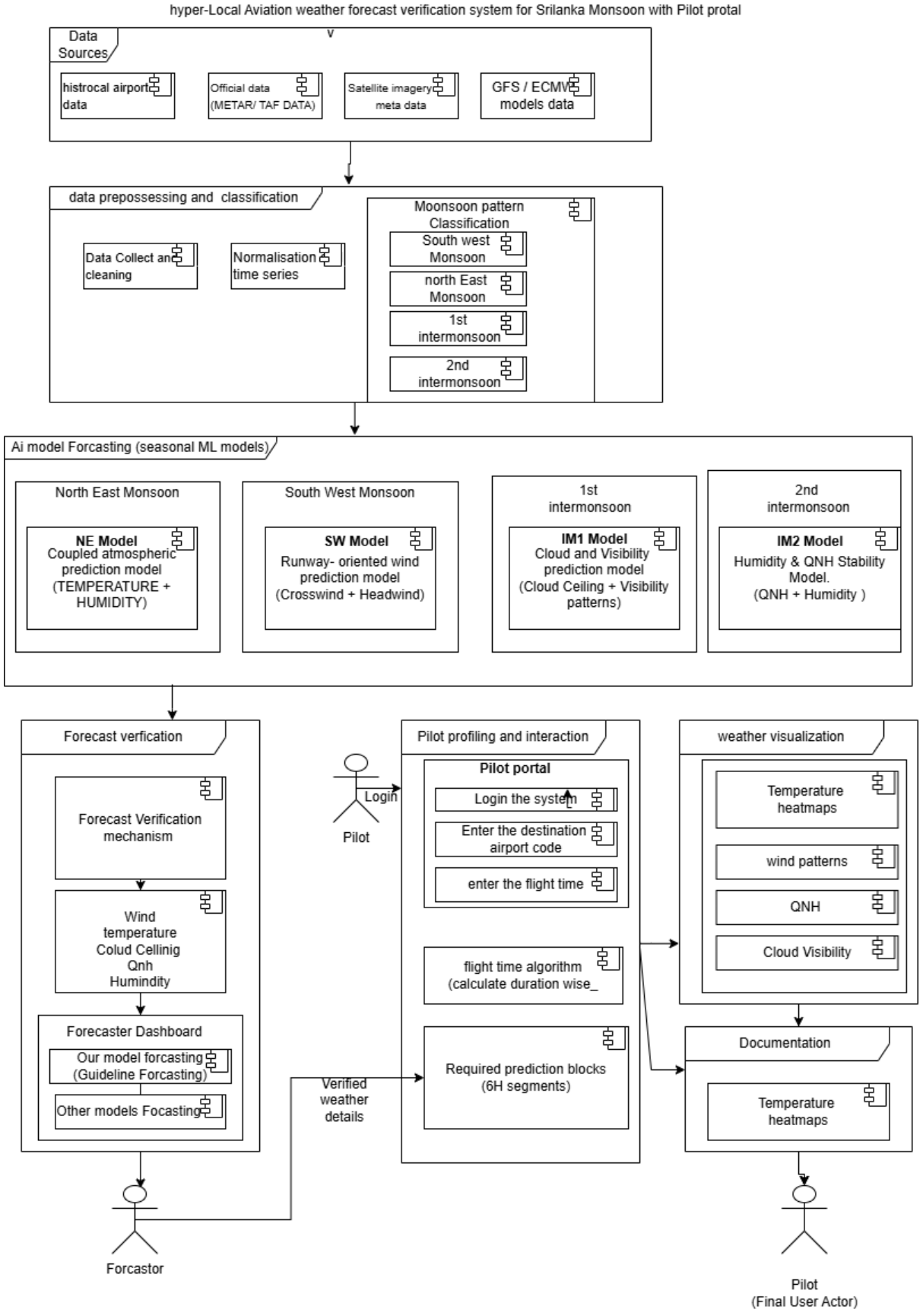


Figure 2-System Structure Diagram

5. USER REQUIREMENTS

5.1 Stakeholder identification

Several stakeholders will interact with or benefit from the proposed weather prediction system.

- **Pilots and Aviation Operators** are the primary users of the system. They require accurate and easily understandable weather forecasts to make safe flight decisions during the Second Inter-Monsoon period.
- **Airport Operational Staff** use weather information to manage flight scheduling, runway operations, and airport safety procedures.
- **Meteorological Analysts** from organizations such as the Department of Meteorology Sri Lanka may use the system to analyze atmospheric trends and validate weather predictions.
- **Researchers and Data Scientists** can use the system to study relationships between atmospheric variables such as humidity, pressure (QNH), and temperature.
- **Airline Management and Aviation Authorities** benefit from improved weather prediction systems that help reduce operational risk and improve flight planning.

5.2 Functional requirements

The system must provide the following functions:

FR1: The system shall collect historical weather data including relative humidity, atmospheric pressure (QNH), and temperature from aviation weather sources.

FR2: The system shall preprocess the collected weather data by handling missing values, cleaning errors, and normalizing time-series datasets.

FR3: The system shall extract relevant atmospheric features such as pressure trends, humidity patterns, and temperature variations.

FR4: The system shall implement a multi-variable machine learning model capable of predicting weather conditions during the Second Inter-Monsoon.

FR5: The system shall generate predictions for humidity and atmospheric pressure simultaneously using a coupled prediction model.

FR6: The system shall calculate a reliability score based on pressure stability and trend accuracy.

FR7: The system shall visualize forecast results using graphical tools such as temperature heatmaps, pressure trend graphs, and wind pattern maps.

FR8: The system shall allow users to view forecast outputs through an interactive dashboard interface.

5.3 Non-functional requirements

The system must satisfy several quality attributes to ensure effective operation.

Security:
The system must ensure secure access to weather data and prediction results. Only authorized users such as pilots and aviation staff should access the dashboard. Sensitive data should be protected using authentication mechanisms.

Performance:
The system should generate predictions within a short processing time so that aviation staff can make timely decisions. Forecast generation should ideally occur within a few seconds after data input.

Usability:
The interface should be simple and intuitive so that pilots and operational staff can easily interpret weather conditions. Visualizations such as heatmaps and trend charts should clearly represent weather patterns.

Reliability:
The system should provide accurate predictions with a high level of consistency. The predictive model should achieve a high trend prediction accuracy and provide reliability scores for each forecast.

Scalability:
The system should support the integration of additional weather stations or datasets in the future without major redesign.

Maintainability:
The system architecture should be modular so that models and data processing components can be updated or improved easily.

6. COMMERCIALIZATION PLAN

The proposed AI-based hyper-local aviation weather forecasting and verification system is developed mainly as a research prototype to demonstrate the use of machine learning techniques in aviation meteorology. The system focuses on analyzing local atmospheric conditions and improving short-term aviation weather prediction using meteorological data from Bandaranaike International Airport.

This project is conducted for academic research purposes and is related to a government domain. Therefore, the developed system is not intended to be commercially sold or used for revenue generation. Business models such as pricing strategies or subscription-based services are not considered within the scope of this study.

Instead, the main purpose of the system is to contribute to public safety and social benefit by supporting more reliable aviation weather forecasting. Accurate weather predictions can help aviation authorities and pilots make safer operational decisions, reducing weather-related risks in flight operations. Therefore, the research aims to provide a technological contribution that supports aviation safety and benefits society.

7. BUDGET AND JUSTIFICATION

- **Hardware Cost**

  Basic hardware resources such as a personal computer or laptop with sufficient processing power and memory are required for data preprocessing, machine learning model development, and system implementation. The project will mainly utilize existing computing devices; however, minor hardware upgrades such as additional storage (SSD or external drive), RAM expansion, or other small accessories may be required to improve processing performance during model training and data analysis.

  Estimated Cost: Rs. 15,000

- **Software and Cloud Services**

  The project will primarily use open-source software tools such as Python, TensorFlow, Scikit-learn, and data visualization libraries for machine learning model development and weather data analysis. In addition, cloud services such as Amazon Web Services (AWS) may be used to store meteorological datasets and perform computational experiments for model training and testing. Since most software tools are open-source, the cost mainly covers cloud service usage and minor development tools.

  Estimated Cost: RS. 20000

- **Processing Resources**

  Additional computational resources may be required to efficiently process large meteorological datasets and run machine learning algorithms. These resources will support data preprocessing, model training, and evaluation tasks required for the aviation weather forecasting system.

  Estimated Cost: Rs. 10,000

Budget Justification

This study is conducted as an academic research project with limited financial requirements. Most of the required resources, including computing devices and development tools, are already available and open-source software will be used for machine learning model development and data visualization. Cloud services such as AWS will only be used on a limited scale for storing meteorological datasets and running computational experiments. Since this research is conducted for academic and experimental purposes and is not intended as a commercial system to be sold to government authorities, the budget allocation remains minimal. The estimated costs mainly support basic computational resources required to conduct and evaluate the proposed aviation weather forecasting system.

8. WORK BREAKDOWN STRUCTURE (WBS)

Task ID	Task Description	Responsible Member	Timeline	Deliverable
1	Literature Review & Problem Identification	All Members	Mar – Apr	Literature review report, problem statement, objectives
2	Proposal Preparation & Submission	All Members	Apr	Approved research proposal
3	Data Collection (Airport Meteorological Data)	All Members	Apr – May	Historical weather data for NE, SW, First & Second Inter-Monsoons
4	Data Preprocessing & Feature Engineering	All Members	May – Jun	Cleaned and processed dataset, features extracted for ML (wind, pressure, humidity, temperature, cloud ceiling, visibility)
5	Development of Pressure-Humidity-Temperature Coupled Prediction Model (NE Monsoon)	Sankadeepa W.D.I (IT22113986)	Jun – Aug	Multi-variable ML model for NE Monsoon, Weather Visualization (temperature heatmaps, wind pattern maps)
6	Development of Wind Pattern Prediction Model (SW Monsoon)	Ranasingha R.A.A.D (IT22269720)	Jun – Aug	Wind Pattern Prediction Model, Runway-Oriented Wind Prediction System, Pilot Portal for flight input and authentication
7	Development of Rapid Weather Change Detection (First Inter-Monsoon)	Jayawardhana H.M.S.I (IT22619976)	Jun – Aug	Rapid Weather Change Detection Model, Flight Time Processing Module, Cloud & Visibility Prediction Model, Threshold Alerts for "Below Minimum"
8	Development of Humidity & QNH Stability Model (Second Inter-Monsoon)	Jayathilaka P.K.V.K (IT22069504)	Jul – Sep	ML model predicting atmospheric stability based on relative humidity and pressure, reliability scoring
9	Weather Document Generation Module	Jayathilaka P.K.V.K (IT22069504)	Sep – Oct	Structured Pilot Guidance Document, Organized Weather Reports for pilots
10	Model Evaluation & Forecast Verification	All Members	Aug – Oct	Verification of accuracy for all models, forecast validation reports
11	Weather Visualization & Dashboard Development	All Members	Sep – Nov	Interactive visualizations for pilots (charts, alerts, dashboards)

12	System Integration, Testing & Result Validation	All Members	Oct – Nov	Fully integrated system, validated predictions and alerts
13	Final Report & Website Development	All Members	Nov – Dec	Final report, project website with dashboard and documentation
14	Final Presentation & Viva	All Members	Dec	Presentation slides, viva-ready system demonstration
15	Research Paper Submission	All Members	Dec	Submitted research paper to journal/conference
16	Research Paper Publication Evidence	All Members	Dec	Proof of publication or acceptance letter

Table 2 - Work breakdown table

9. GANTT CHART

Task	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Literature Review & Problem identification										
Proposal Preparation & Submission										
Data Collection (Airport Meteorological Data)										
Data Preprocessing & Feature Engineering										
Development of Coupled Prediction Model										
Model Evaluation & Forecast verification										
Weather Visualization Development										
System Testing & Result Validation										
Final Report & Website Development										
Final Presentation & Viva										
Research Paper Submission										
Research Paper Publication Evidence										

Table 3 - project Implementation Timeline (Gantt Chart)

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11. APPENDICES

11.1 Survey Instruments

This research does not involve surveys or questionnaires because the study focuses on meteorological data analysis and machine learning–based weather prediction. Therefore, no survey instruments are required for this project Detailed diagrams.

11.2 Preliminary UI mockups

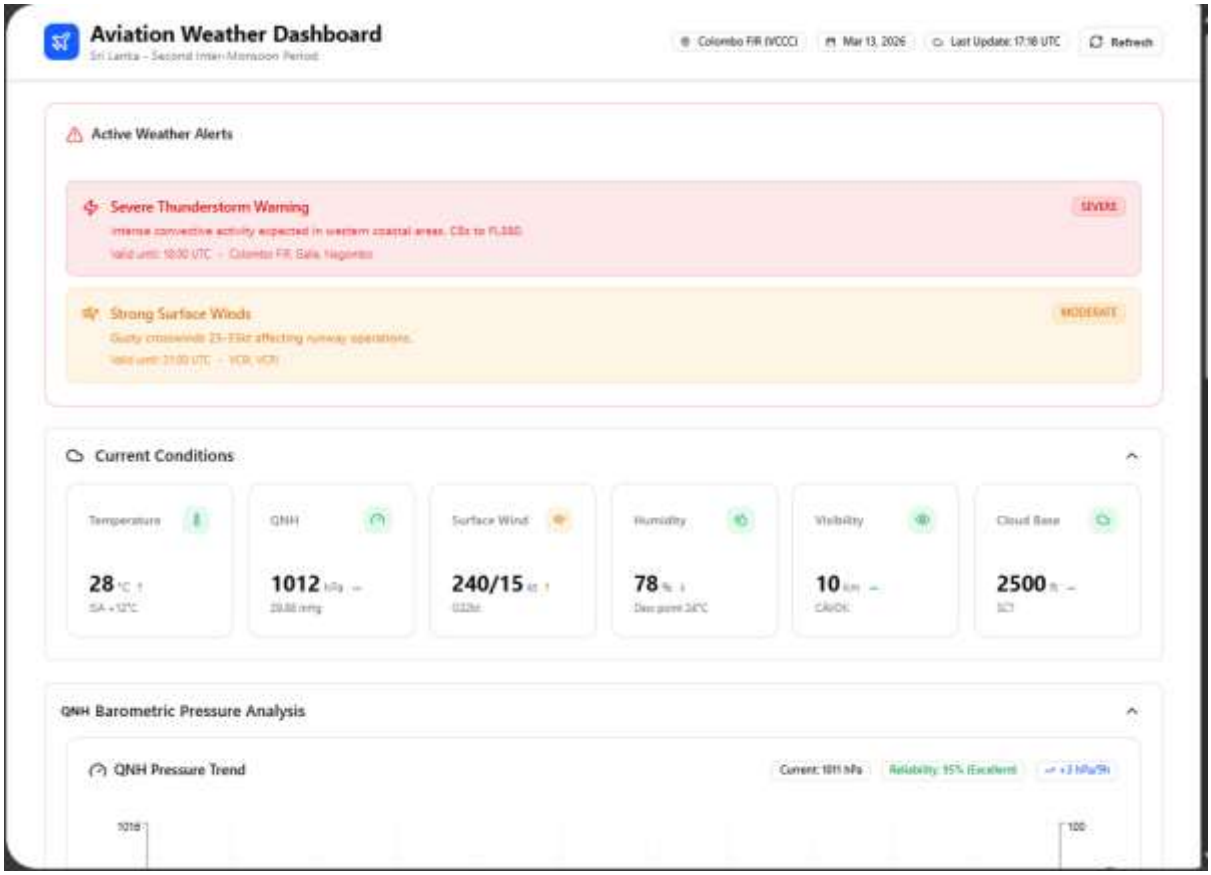


Figure 3-UI mockup

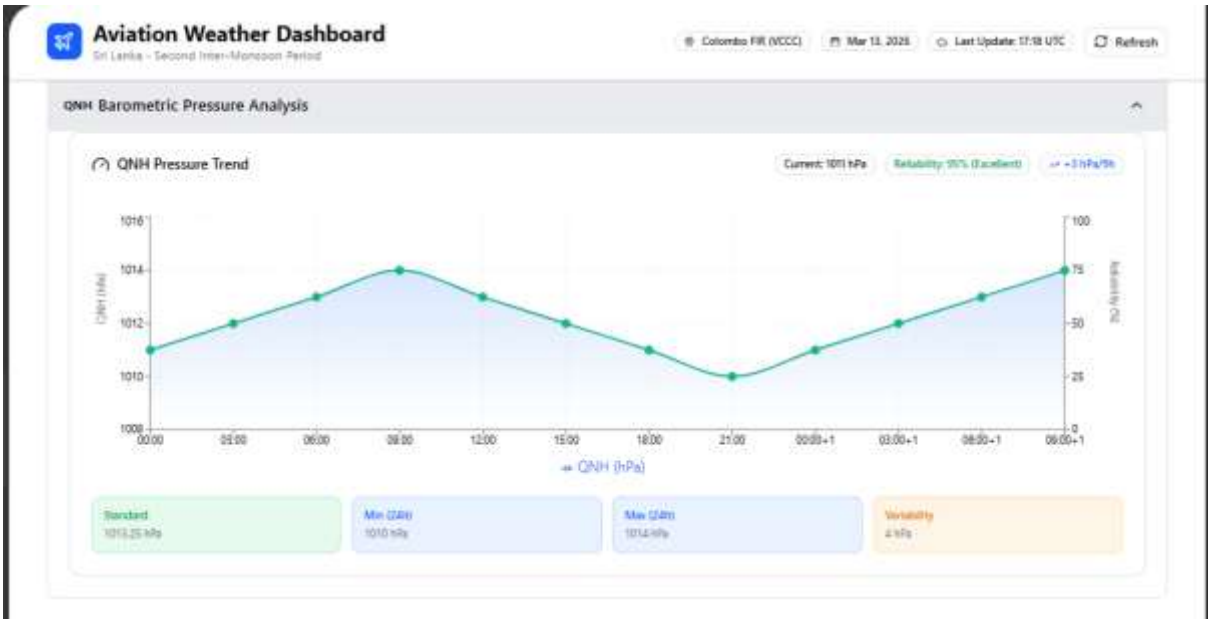


Figure 4-UI Mockup

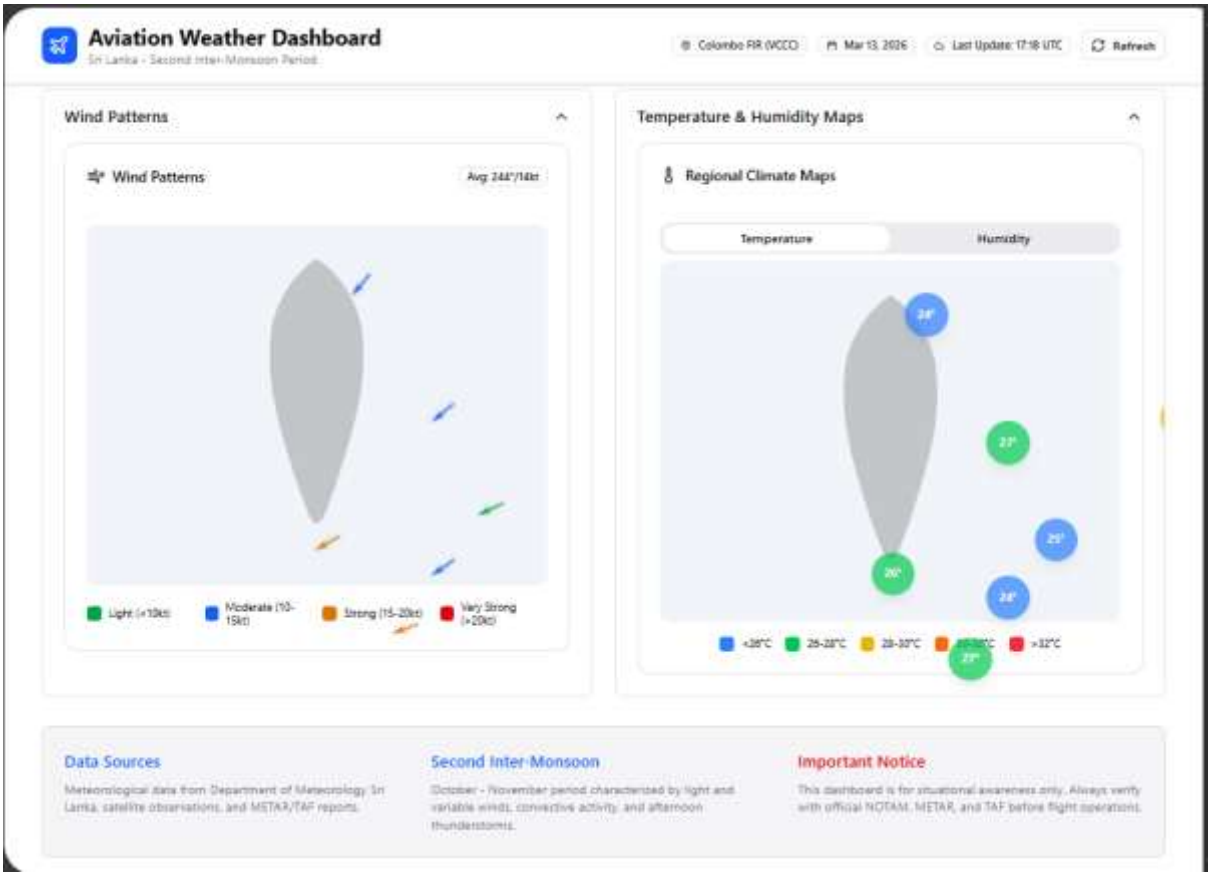


Figure 5-UI Mockup

11.3 Risk Analysis

Risk ID	Risk Description	Probability	Impact	Mitigation Strategy
R1	Limited availability or incomplete historical weather data from Bandaranaike International Airport	Medium	High	Collect data from multiple reliable sources such as the Department of Meteorology and aviation databases. Apply data preprocessing and missing value handling techniques.
R2	Poor prediction accuracy of the machine learning model due to insufficient training data	Medium	High	Use data augmentation techniques and optimize model parameters. Evaluate multiple algorithms such as ANN and LSTM to select the best-performing model.
R3	Difficulty in capturing sudden extreme weather events during the Second Inter-Monsoon	Medium	Medium	Include historical extreme weather events in training datasets and apply hybrid data augmentation strategies.
R4	Complexity in integrating multiple atmospheric variables (humidity, QNH, temperature) into a single predictive model	Low	Medium	Use a coupled multi-variable machine learning model and conduct feature selection and feature engineering.
R5	Visualization tools may not clearly communicate weather conditions to pilots	Low	Medium	Design pilot-friendly dashboards using simple charts, heatmaps, and wind pattern maps with clear color coding.
R6	Technical issues related to software tools or programming environment	Low	Medium	Use well-supported tools such as Python, TensorFlow, and visualization libraries with proper documentation and version control.
R7	Time constraints in data preprocessing, model training, and evaluation	Medium	Medium	Follow a structured project timeline and divide tasks clearly among team members.
R8	Ethical or data usage restrictions related to meteorological datasets	Low	Low	Obtain permission from relevant authorities and ensure proper citation and responsible data usage.

Table 4 - Risk Analysis